

CyberShield AI – Behavioral Intrusion Detection System

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Problem Statement

- Cyber attacks are rapidly increasing in both frequency and complexity across digital systems.
- Traditional intrusion detection systems rely on known attack signatures and fail to detect unknown threats.
- Zero-day attacks remain undetected due to lack of intelligent analysis.
- There is a strong need for real-time, adaptive, and intelligent security solutions.

Objective

- The main objective is to detect anomalous network behavior using machine learning techniques.
- The system aims to provide real-time monitoring and alert mechanisms.
- It focuses on improving cybersecurity awareness through visualization.
- The project also enables interpretable AI decisions for better understanding.

Proposed Solution

- AI-Driven Anomaly Detection: Utilizes an unsupervised machine learning model (Isolation Forest) to proactively identify both known attack vectors and novel, zero-day network threats.
- Interactive Analytics Dashboard: A secure, centralized web portal featuring dynamic data visualization to monitor traffic trends, attack distributions, and overall network health.
- Real-Time Threat Alerting: Immediately triggers severity-tiered visual notifications (Low, Medium, High) the moment malicious packets are processed or simulated.
- Explainable AI (XAI) Integration: Transcends "black-box" machine learning by providing human-readable justifications (e.g., "suspicious payload ratio") for every flagged network event.

System Overview

- Flexible Data Ingestion: Supports both manual packet injection for targeted analysis and bulk CSV uploads for heavy-duty batch processing.
- AI-Driven Threat Detection: An Isolation Forest machine learning model scans core network features to identify zero-day anomalies and known attack vectors.
- Explainable AI (XAI) & Scoring: Automatically calculates model confidence, assigns a severity tier (Low/Medium/High), and provides a human-readable reason for the flag.
- Real-Time Visualization & Alerts: Pushes processed data to an interactive analytics dashboard, triggering instant UI popup warnings for high-severity threats.

Technologies Used

- Python with Flask framework is used for backend development.
- Scikit-learn is used for implementing machine learning algorithms.
- HTML, CSS, and JavaScript are used for frontend development.
- SQLite is used as the database.
- Chart.js is used for data visualization.
- The ReportLab library is used to generate downloadable, formatted PDF incident reports on-the-fly.
- Pyotp is used to handle the Time-Based One-Time Password (TOTP) logic.
- Qrcode is used to generate the QR code image that users scan with an authenticator app (like Google Authenticator or Authy) to link their account.

Machine Learning Model

- Core Engine: Powered by the Isolation Forest algorithm for robust anomaly detection.
- Unsupervised Learning: Requires no pre-labeled data, allowing the system to identify novel, zero-day threats naturally.
- Detection Mechanism: Flags malicious traffic by mathematically isolating statistical outliers from normal network patterns.
- High Performance: Highly efficient and inherently scalable, making it ideal for processing large volumes of network packets.

Features

- Live Threat Detection: Real-time monitoring with instant, severity-based alerts.
- Dual-Mode Ingestion: Seamlessly handles both manual packet testing and bulk CSV uploads.
- Interactive Dashboard: Dynamic, multi-tenant portal for live data visualization and trend analysis.
- Automated Reporting: On-the-fly PDF generation for comprehensive incident logging.

Authentication

- Secure User Onboarding: Isolated database environments (SQLite) ensure multi-tenant data privacy and prevent duplicate registrations.
- Cryptographic Password Hashing: Zero plain-text storage; utilizes industry-standard Werkzeug security for cryptographic salting and hashing.
- Two-Factor Authentication (2FA): Mandatory TOTP (Time-Based One-Time Password) integration via QR codes for Google Authenticator/Authy.
- Strict Session Management: Implementation of route-guarding (@login_required) to actively intercept and block unauthorized dashboard access.

Dashboard

- The dashboard displays total network packets analyzed.
- It shows classification between normal and attack traffic.
- Severity levels are indicated for detected threats.
- Graphs and charts provide visual insights into data.

Output Screenshots



CYBERSHIELD AI

Behavioral Intrusion Detection System

Username

Password

Secure Login

Don't have an account? [Register here.](#)



TWO-FACTOR AUTH

Enter the 6-digit code from your app

Authenticate



CYBERSHIELD AI

Scan Engine

Analytics Dashboard

Logout

Dark Mode ☒

Threat Detection Engine

Analyze network packets manually or via batch CSV upload.

Manual Packet Injection

Duration (ms):

0

Source Bytes:

250

Destination Bytes:

3000

Connection Count:

5

Service Count:

5

Analyze Packet

Batch Analysis (CSV)

Fields: duration, src_bytes, dst_bytes, count, srv_count

Choose File No file chosen

Process Batch File



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Dark Mode ☒

Manual Packet Injection

Duration (ms):

0

Source Bytes:

250

Destination Bytes:

3000

Connection Count:

5

Service Count:

5

Analyze Packet

Prediction: Normal

Confidence: 62.54%

Severity: Low

Type: None

XAI Analysis: Traffic within expected parameters

Batch Analysis (CSV)

Fields: duration, src_bytes, dst_bytes, count, srv_count

Choose File sample.csv

Process Batch File

Batch Processed. Attacks: 4 | Normal: 1

View Analytics Dashboard



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Traffic Analytics & XAI

▶ Start Live Simulation

↓ Download PDF Report

Total Packets

0

Normal Traffic

0

Threats Detected

0

Model Architecture

Algorithm: Isolation Forest
Dataset Base: NSL-KDD Synthetic
Core Features: 5 (Dur, Src/Dst Bytes, Counts)
XAI Logic: Rule-based thresholds

Traffic Distribution

Attack Trend Over Time

1



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Scan Engine

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Dark Mode ☒

Traffic Distribution

Attack Trend Over Time

1

0

Normal Attack

Live Prediction Feed & Analysis

Time	Status	Confidence	Severity	Threat Type	XAI Reason
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No recent data.



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Scan Engine

Analytics Dashboard

Logout

Dark Mode ☒



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Traffic Analytics & XAI

[▶ Start Live Simulation](#) [↓ Download PDF Report](#)

Total Packets

6

Normal Traffic

2

Threats Detected

4

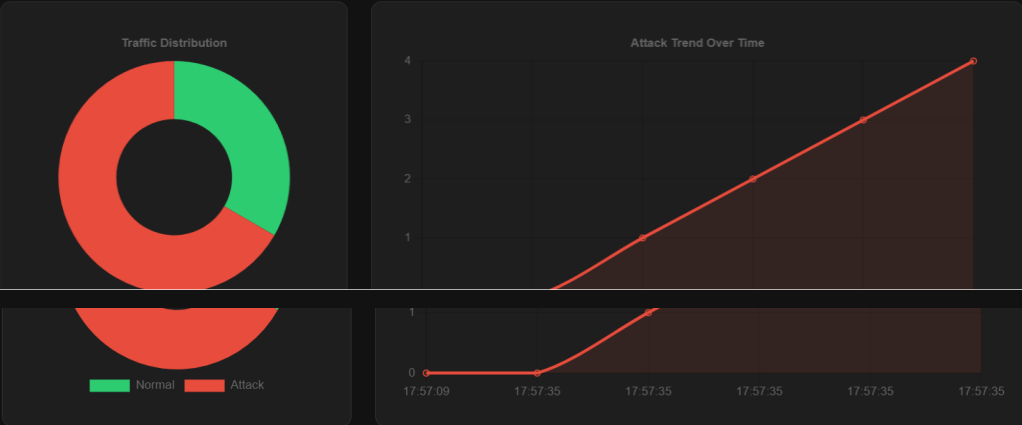
Model Architecture

Algorithm: Isolation Forest

Dataset Base: NSL-KDD Synthetic

Core Features: 5 (Dur, Src/Dst Bytes, Counts)

XAI Logic: Rule-based thresholds



Live Prediction Feed & Analysis

Time	Status	Confidence	Severity	Threat Type	XAI Reason
17:57:35	Attack	53.98%	Low	U2R	Anomalous multivariate pattern detected
17:57:35	Attack	68.76%	High	Probe	Abnormal duration, High source bytes
17:57:35	Attack	58.31%	Low	U2R	Anomalous multivariate pattern detected
17:57:35	Attack	64.45%	Low	U2R	Anomalous multivariate pattern detected
17:57:35	Normal	62.54%	Low	None	Traffic within expected parameters.
17:57:09	Normal	62.54%	Low	None	Traffic within expected parameters.



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Scan Engine

Analytics Dashboard

Logout

Dark Mode ☒

Traffic Analytics & XAI

Total Packets
11

Normal Traffic
4

Attacks
7

Threat Detected: U2R
Severity: Low | Conf: 50.38%
Reason: Anomalous multivariate pattern detected

Threat Detected: Probe
Severity: High | Conf: 60.57%
Reason: Abnormal duration, High source bytes, Unusual connection count, Suspicious payload ratio

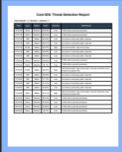
Threat Detected: U2R
Severity: Low | Conf: 50.64%
Reason: Anomalous multivariate pattern detected

Traffic Distribution

Attack Trend Over Time

(anonymous)

1 / 1 | 100% | [Icons]



1

Core-IDS: Threat Detection Report

Total Packets: 19 | Normal: 8 | Attacks: 11

Time	Type	Status	Conf.	Severity	XAI Reason
17:57:09	None	Normal	62.54%	Low	Traffic within expected parameters.
17:57:35	None	Normal	62.54%	Low	Traffic within expected parameters.
17:57:35	U2R	Attack	64.45%	Low	Anomalous multivariate pattern detected
17:57:35	U2R	Attack	58.31%	Low	Anomalous multivariate pattern detected
17:57:35	Probe	Attack	68.76%	High	Abnormal duration, High source bytes
17:57:35	U2R	Attack	53.98%	Low	Anomalous multivariate pattern detected
17:59:39	U2R	Attack	50.64%	Low	Anomalous multivariate pattern detected
17:59:42	None	Normal	50.04%	Low	Traffic within expected parameters.
17:59:44	None	Normal	60.1%	Low	Traffic within expected parameters.
17:59:47	Probe	Attack	60.57%	High	Abnormal duration, High source bytes, Unusual connection count, Suspicious payload ratio
17:59:49	U2R	Attack	50.38%	Low	Anomalous multivariate pattern detected
17:59:52	U2R	Attack	51.14%	Low	Anomalous multivariate pattern detected
17:59:54	U2R	Attack	51.41%	Low	Anomalous multivariate pattern detected
17:59:57	U2R	Attack	51.68%	Low	Anomalous multivariate pattern detected

Advantages

- The system provides real-time intrusion detection.
- It uses AI for intelligent threat detection.
- The interface is user-friendly and easy to use.
- Explainable AI improves transparency and trust.

Limitations

- The system currently uses synthetic datasets.
- It has a limited number of features.
- It is not fully deployed on real-world networks.
- Performance may vary with different datasets.

Future Enhancements

- Deep learning models can be integrated for improved accuracy.
- Real network traffic data can be incorporated.
- The system can be deployed on cloud platforms.
- A mobile application interface can be developed.

Conclusion

- CyberShield AI is an intelligent intrusion detection system.
- It combines artificial intelligence with cybersecurity.
- It enhances the detection of modern cyber threats.
- The system has strong potential for real-world applications.